Snowflake Interview Q&A; - Complete Cheat Sheet

From Basic to Advanced | Data Engineer / Analyst / Admin Ready | 2026 Edition

LEVEL 1: BASICS - Must Know

1. What is Snowflake? Why is it different?

Cloud-native SaaS data platform on AWS/Azure/GCP. Key differentiators: Separation of Storage & Compute, Zero-copy cloning, Time Travel, Auto-scaling, Pay-per-second, No indexes/vacuuming needed.

2. Explain Snowflake Architecture - 3 Layers

Storage Layer: Columnar micro-partitions (50-500MB) stored on S3/Azure/GCS.
Compute Layer: Virtual Warehouses - independent MPP clusters.
Cloud Services Layer: Metadata, optimizer, security, auth - serverless & auto-scaled by Snowflake.

3. What is a Micro-partition?

Atomic storage unit, ~50-500MB compressed, columnar, immutable. Auto-created. Contains metadata: min/max, distinct values, etc. for pruning.

4. What is a Virtual Warehouse? Sizes?

MPP compute cluster. Sizes: XS, S, M, L, XL, 2XL...6XL. Each step doubles credits & compute. Features: auto-suspend, auto-resume, auto-scale (multi-cluster).

5. Types of Tables in Snowflake?

Permanent (90 days Time Travel + 7 days Fail-safe), Transient (0-90 days TT, no Fail-safe, cheaper), Temporary (session-bound, 0-1 day TT, no Fail-safe), External.

6. Transient vs Temporary?

Transient: For staging/transient data, cheaper, no Fail-safe. Temporary: Deleted after session, only for personal temp work.

7. What is Time Travel and Fail-safe?

Time Travel: Query historical data using AT/OFFSET/BEFORE. 1 day default for all, up to 90 days for Enterprise permanent tables.
Fail-safe: 7-day disaster recovery by Snowflake after Time Travel expires. Not user-accessible.

LEVEL 2: INTERMEDIATE - Core Features

8. What is Zero-Copy Cloning?

Instant metadata-only clone. No data copy. **Syntax:** CREATE TABLE new_tbl CLONE old_tbl AT(OFFSET => -60*5) . Works for DB/Schema/Table. Great for dev/test.

9. Snowpipe vs COPY command?

COPY: Bulk batch load, you manage warehouse.
Snowpipe: Serverless continuous micro-batch, auto-ingest via S3 Event Notifications + SQS. Use Snowpipe for near real-time.

10. What are Streams and Tasks?

Stream: CDC object that tracks DML (INSERT/UPDATE/DELETE) - shows change metadata like METADATA$ACTION, METADATA$ISUPDATE.
Task: Cron-like scheduler (USING CRON). Stream + Task + MERGE = serverless ELT pipeline.

11. What is a Stage? Types?

Location for files before load/after unload.
Types: User Stage (@~), Table Stage (@%table_name), Named Internal Stage, External Stage (S3/GCS/Azure URL).

12. Explain Clustering

For tables >1TB. Organizes micro-partitions by clustering key to improve pruning. **Syntax:** CREATE TABLE ... CLUSTER BY (col1, col2). Check clustering: SYSTEM$CLUSTERING_INFORMATION('table'). Auto-clustering in background (costs credits).

13. Caching in Snowflake?

1. Result Cache: 24h in Cloud Services - same query re-run returns instantly if data unchanged.
2. Metadata Cache: Micro-partition pruning metadata.
3. Warehouse Cache (Local Disk): SSD cache on warehouse nodes - persists until warehouse suspended.

14. Secure View vs Materialized View?

Secure View: Hides DDL definition from users, needed for data privacy.
Materialized View: Pre-computed, auto-maintained. Speeds up queries. Only Enterprise+.

15. How does Snowflake handle semi-structured data?

Native VARIANT, OBJECT, ARRAY types. Supports JSON, Parquet, Avro, ORC, XML. Query with dot notation: src:city , and FLATTEN() to explode arrays. Best practice: Use COPY with FILE_FORMAT.

LEVEL 3: ADVANCED - Performance, Security, Architecture

16. What is Search Optimization Service?

For highly selective point lookups (WHERE id = ?). Creates search access path. Ideal for OLTP-like queries. Enable: ALTER TABLE ... ADD SEARCH OPTIMIZATION ON EQUALITY(id).

17. How do you optimize performance?

1. Right-size warehouse 2. Enable clustering if clustering_depth is high 3. Use Search Optimization 4. Leverage result cache 5. Avoid SELECT * 6. Use Materialized Views 7. Analyze Query Profile for spillage/pruning 8. Use QUALIFY to filter window functions.

18. Data Sharing / Secure Sharing?

Share live data without copying. Provider creates SHARE, adds objects, grants to consumer account. Consumer creates DB FROM SHARE. Now enhanced as Snowflake Marketplace Listings. Reader Account for non-Snowflake customers.

19. Explain RBAC

Role-Based Access Control. Roles -> Users. Hierarchy: ACCOUNTADMIN > SECURITYADMIN > SYSADMIN > custom roles. Privileges: USAGE, SELECT, OWNERSHIP etc. Always use ROLE inheritance and future grants: GRANT SELECT ON FUTURE TABLES IN SCHEMA...

20. Row-Level & Column-Level Security?

Row Access Policy: CREATE ROW ACCESS POLICY ... Controls which rows user sees.
Masking Policy (Dynamic Data Masking): CREATE MASKING POLICY ... masks column values (e.g., show only last 4 digits of SSN).

21. What is Snowpark?

Developer framework to write Python/Java/Scala directly in Snowflake, pushing compute to warehouse. Use Session, DataFrame API. No need for external Spark. Supports UDFs, UDTFs, Stored Procedures.

22. External Tables vs Internal Tables?

External Tables: Query files in S3 directly without loading. Slower, but good for Data Lake. Refresh via ALTER EXTERNAL TABLE ... REFRESH.
Internal: Loaded micro-partitions, fast.

23. Warehouse auto-suspend & auto-resume?

auto_suspend = 60 (seconds of inactivity). Saves credits. auto_resume = TRUE resumes on new query. Set low for dev (60s), higher for prod (300-600s).

24. What is Cost Control?

Use RESOURCE MONITORS to set credit quotas, STATEMENT_TIMEOUT_IN_SECONDS, STATEMENT_QUEUED_TIMEOUT_IN_SECONDS, Limit warehouse sizes, Use transient tables, Monitor via ACCOUNT_USAGE schema.

LEVEL 4: SCENARIO-BASED - Where Interviews Are Decided

25. Query slow on 5TB table - Debug?

Steps: 1. Open Query Profile 2. Check % scanned vs pruned 3. Check for Spilling to remote/local disk (needs bigger warehouse) 4. Check clustering_depth via SYSTEM$CLUSTERING_INFORMATION 5. Check if using cache 6. Use EXPLAIN to see plan.

26. How to handle SCD Type 2?

Use Stream on source table -> Task every 5 min -> MERGE into Dim: When MATCHED and hash diff -> UPDATE end_date = current_date + INSERT new row. When NOT MATCHED -> INSERT. Use HASH(*) for change detection.

27. COPY failed for 2 files out of 1000?

Use COPY_HISTORY and VALIDATION_MODE = RETURN_ERRORS. Query: SELECT * FROM INFORMATION_SCHEMA.LOAD_HISTORY or SELECT * FROM TABLE(VALIDATE(table_name, JOB_ID => '_last')). Also check @stage file status.

28. UNION vs UNION ALL in Snowflake?

UNION does DISTINCT sort which is expensive on large data. Always prefer UNION ALL unless de-duplication is required.

29. Concurrency handling?

Use Multi-cluster warehouses (min=1, max=3) with scaling policy STANDARD (immediate) or ECONOMY (saves credits). Prevents queuing.

30. Key SQL Functions to know

QUALIFY (filter window functions), FLATTEN + LATERAL, IFF (inline IF), ZEROIFNULL, DECODE, DATE_TRUNC, DATEDIFF, HASH, TRY_CAST, OBJECT_CONSTRUCT.

31. Bonus: What interviewers love to hear

Mention: 'I built ELT: S3 Event -> Snowpipe -> Stream captures CDC -> Task runs MERGE every 5 mins -> Fact table with Clustering on date_id + Masking policy for PII.' + Monitored cost with Resource Monitor + Used Query Profile to tune.

Quick Revision Formula: Micro-partitions + 3 Layers (Storage/Compute/Services) + Zero-Copy + Time Travel + Stream/Task + Clustering + RBAC + Cost Control. Be ready to whiteboard a pipeline!